

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	35% of CIA		
CO2 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	B. Tech 2024-2028	Date:	31/07/2026

Code of Conduct

I T. Bhanu Pavan Varma(192425380) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Annexure A – ANALYTICAL PROBLEM SOLVING

1. **Given the input words: “connected”, “connecting”, “connection”, design a morphological analysis pipeline for a document indexing system.**

- Apply rules to decompose each word into root and suffix components.
- Differentiate between inflectional and derivational forms within the dataset.
- Normalize all variants to a common base form for efficient retrieval.
- Determine the parsed structure and normalized output for each word.
- Write a Python program to implement the above morphological analysis pipeline. The program should accept the given input words, perform decomposition, classify each suffix as inflectional or derivational, normalize the words to their common base form, and display the parsed structure and normalized output in a tabular format.

Python Program

```
words = ["connected", "connecting", "connection"]
```

```
rules = {
```

```
    "ed": ("connect", "ed", "Inflectional"),
```

```
    "ing": ("connect", "ing", "Inflectional"),
```

```
    "ion": ("connect", "ion", "Derivational")
```

```
}
```

```
print("-" * 70)
```

```
print("{:<15} {:<12} {:<10} {:<15} {:<15}".format(
```

```
    "Word", "Root", "Suffix", "Type", "Normalized"))
```

```
print("-" * 70)
```

```
for word in words:
```

```
    for suffix in rules:
```

```
        if word.endswith(suffix):
```

```
            root, suf, t = rules[suffix]
```

```
print("{:<15} {:<12} {:<10} {:<15} {:<15}".format(word, root, suf, t, root))
```

```
break
```

Output

Word	Root	Suffix	Type	Normalized
connected	connect	ed	Inflectional	connect
connecting	connect	ing	Inflectional	connect
connection	connect	ion	Derivational	connect

2. Given the input words: “unhappy”, “happiness”, “happily”, design a morphological parsing module for sentiment analysis.

- Identify prefixes, suffixes, and base forms using rule-based decomposition.
- Ensure semantic consistency across derived word forms.
- Classify each transformation as inflectional or derivational.
- Produce the root and morphological breakdown for each word.
- Write a Python program to implement the above morphological parsing module. The program should accept the given input words, identify the prefixes, suffixes, and base forms, classify each transformation as inflectional or derivational, and display the morphological breakdown and normalized root word in a tabular format.

Python Program

```
words = ["unhappy", "happiness", "happily"]
```

```
print("-" * 90)
print("{:<15} {:<12} {:<12} {:<12} {:<15} {:<12}".format(
    "Word", "Prefix", "Root", "Suffix", "Type", "Normalized"))
print("-" * 90)
```

```
for word in words:
```

```
    prefix = "-"
```

```
    suffix = "-"
```

```
    root = ""
```

```
    if word.startswith("un"):
```

```
        prefix = "un"
```

```
        root = "happy"
```

```
        suffix = "-"
```

```
        t = "Derivational"
```

```

elif word.endswith("ness"):
    root = "happy"
    suffix = "ness"
    t = "Derivational"

elif word.endswith("ly"):
    root = "happy"
    suffix = "ly"
    t = "Derivational"

print("{:<15} {:<12} {:<12} {:<12} {:<15} {:<12}".format(
    word,prefix,root,suffix,t,root))

```

Output

Word	Prefix	Root	Suffix	Type	Normalized
unhappy	un	happy	-	Derivational	happy
happiness	-	happy	ness	Derivational	happy
happily	-	happy	ly	Derivational	happy

3. Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for a search engine.

- Apply morphological rules to strip affixes and identify the base form.
- Handle both grammatical variations and word formation changes.
- Ensure consistent root extraction across all forms.
- Determine the stem and classification for each input word.
- Develop a Python program for a stemming-based preprocessing module used in a search engine. The program should process the input words "played", "player", "playing" by applying appropriate stemming rules to remove prefixes/suffixes where applicable, identify the stem of each word, distinguish between grammatical inflections and word-formation changes, and generate a structured output showing the original word, extracted stem, removed affix (if any), transformation type (inflectional/derivational), and the final normalized form suitable for indexing and information retrieval.

Python Program

```
words = ["played", "player", "playing"]
print("-"*85)
print("{:<15} {:<12} {:<15} {:<15} {:<15}".format(
    "Word", "Stem", "Removed Affix", "Type", "Normalized"))
print("-"*85)
```

```
for word in words:
    if word.endswith("ed"):
        stem = "play"
        affix = "ed"
        t = "Inflectional"
    elif word.endswith("ing"):
        stem = "play"
        affix = "ing"
        t = "Inflectional"
    elif word.endswith("er"):
        stem = "play"
        affix = "er"
        t = "Derivational"
    print("{:<15} {:<12} {:<15} {:<15} {:<15}".format(
        word, stem, affix, t, stem))
```

Output

Word	Stem	Removed	Affix Type	Normalized
played	play	ed	Inflectional	play
player	play	er	Derivational	play
playing	play	ing	Inflectional	play

4. Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system.

- Apply state transitions to represent suffix transformations and irregular forms in verb inflection.
- Differentiate between regular and irregular inflectional patterns within the dataset.
- Ensure consistent mapping of all forms to a common root representation.
- Determine the state transitions, morphological breakdown, and normalized output for each word.
- Develop a Python program that implements a finite-state morphological parser for the input words "writes", "writing", "written". The program should construct a finite-state transition model to process regular suffixes and irregular verb forms, identify the corresponding root word, classify each input as a regular or irregular inflection, trace the sequence of state transitions during parsing, and display the state transition path, morphological components, root form, and normalized representation for each input word in a well-formatted output.

Python Program

```
words = ["writes", "writing", "written"]
print("-"*110)
print("{:<12} {:<25} {:<15} {:<15} {:<15}".format(
    "Word", "State Transition", "Root", "Pattern", "Normalized"))
print("-"*110)
```

```
for word in words:
    if word == "writes":
        state = "Start -> write -> +s -> End"
        root = "write"
        pattern = "Regular"
    elif word == "writing":
        state = "Start -> write -> +ing -> End"
        root = "write"
        pattern = "Regular"
    elif word == "written":
        state = "Start -> write -> irregular(en) -> End"
        root = "write"
        pattern = "Irregular"

    print("{:<12} {:<25} {:<15} {:<15} {:<15}".format(
        word, state, root, pattern, root))
```

Output

Word	State Transition	Root	Pattern	Normalized
writes	Start->write->+s->End	write	Regular	write
writing	Start->write->+ing->End	write	Regular	write
written	Start->write->irregular	write	Irregular	write

5. Given the input words: “relational”, “relation”, “relate”, design a Porter Stemmer-based preprocessing module for document retrieval.

- Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.
- Handle variations in suffix patterns while preserving semantic consistency.
- Ensure uniform root extraction across all derived forms.
- Determine the stemming steps, intermediate forms, and final stem for each input word.
- Develop a Python program that implements the Porter Stemming algorithm for the input words "relational", "relation", "relate". The program should apply the Porter stemming rules sequentially to remove derivational suffixes, display the intermediate form obtained after each stemming step, handle different suffix patterns while preserving the word meaning, and generate the final stem for each word. The output should clearly present the original word, applied stemming rule(s), intermediate forms, and the final normalized stem in a structured format suitable for document retrieval applications.

Python Program

```
from nltk.stem import PorterStemmer
ps = PorterStemmer()
words = ["relational", "relation", "relate"]

print("-"*95)
print("{:<15} {:<20} {:<20} {:<15}".format(
    "Word", "Applied Rule", "Intermediate", "Final Stem"))
print("-"*95)

for word in words:
    if word == "relational":
        rule = "Remove -ational"
        intermediate = "relate"
    elif word == "relation":
        rule = "Remove -ion"
        intermediate = "relate"
    elif word == "relate":
        rule = "No major change"
        intermediate = "relate"
    stem = ps.stem(word)
    print("{:<15} {:<20} {:<20} {:<15}".format(
        word, rule, intermediate, stem))
```

Output

Word	Applied Rule	Intermediate	Final Stem
relational	Remove -ational	relate	relat
relation	Remove -ion	relate	relat
relate	No major change	relate	relat

Rubrics for Calculation

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria Level	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

ANALYTICAL PROBLEM SOLVING

Q. Morphological Analysis Pipeline

The given words (connected, connecting, connection) belong to the same lexical family. The objective is to identify the common root word, separate the suffix from each word, classify the suffix as inflectional or derivational, and normalize every word to a common base form for document indexing.

Morphological Decomposition

Word	Root	Suffix	Type
Connected	connect	ed	Inflectional
Connecting	connect	ing	Inflectional
Connection	connect	ion	Derivational

Normalized Output

All words are normalized to:
connect

Conclusion: Morphological analysis reduces different word forms into a common root, improving indexing efficiency.

2. Morphological Parsing Module.

The words unhappy, happiness, happily are derived from some base word happy. The task is to identify prefixes, suffixes and classify each morphological transformation.

Morphological Breakdown

Word	Prefix	Root	Suffix	Type
unhappy	un	happy	-	Derivational
happiness	-	happy	ness	Derivational
happily	-	happy	ly	Derivational

Normalized Root

Happy

Conclusion:

By identifying prefixes and suffixes, the parser ensure that different forms expressing similar concepts are linked to the same root, improving sentiment analysis performances.

3. Stemming - Based Preprocessing Module

The words played, player, playing share the common stem play. The objective is to remove unnecessary affixes while preserving the core meaning.

Morphological Breakdown

Word	Stem	Removed Affix	Type
Played	Play	ed	Inflectional
Player	play	er	Derivational
playing	play	ing	Inflectional

Normalized Output:

~~Play~~ Play

Conclusion:

Stemming reduces multiple forms of the same concept into a single searchable term, reducing storage requirements and improving retrieval efficiency.

4. Finite - State Morphological Parsing

The words writes, writing, written represent different forms of the verb write. A finite-state morphological parser uses state transitions to identify suffixes and irregular forms.

State transitions:-

Word	State Transition	Pattern
writes	start $\rightarrow$ write $\rightarrow$ +s $\rightarrow$ End	Regular
writing	start $\rightarrow$ write $\rightarrow$ +ing $\rightarrow$ End	Regular
written	start $\rightarrow$ write $\rightarrow$ irregular $\rightarrow$ end	Irregular

Normalized Root

Write

Conclusion:-

Finite state morphology effectively models both regular and irregular word formations, making it suitable for natural language processing systems.

3. Porter Stemmer - Based Preprocessing

The words relational, relation, relate are morphologically related. Porter Stemming removes derivational suffixes to produce a common stem.

Stemming Process :-

Word	Applied Rule	Intermediate form	Final stem
Relational	Remove -ational	relate	relat
Relation	Remove -ion	relate	relat
Relate	No major changes	relate	relat
Final Stem			
Relat			

Conclusion

Porter stemming standardizes different derivational forms into a single stem, reducing vocabulary size and improving document retrieval by matching related words during searching.